FABRICATION AND PHYSICOCHEMICAL CHARACTERIZATION OF PVA/GUAR GUM NANOFIBERS AS FAST DISSOLVED SCAFFOLD FOR DERMIS REGENERATION

Amr Ahmed Ismail[1], Dr. Norjihada Izzah Ismail[1]

Department of Biomedical Engineering and Health Sciences, Faculty of Electrical Engineering, Universiti Teknologi Malaysia, 81310, Johor, Malaysia.

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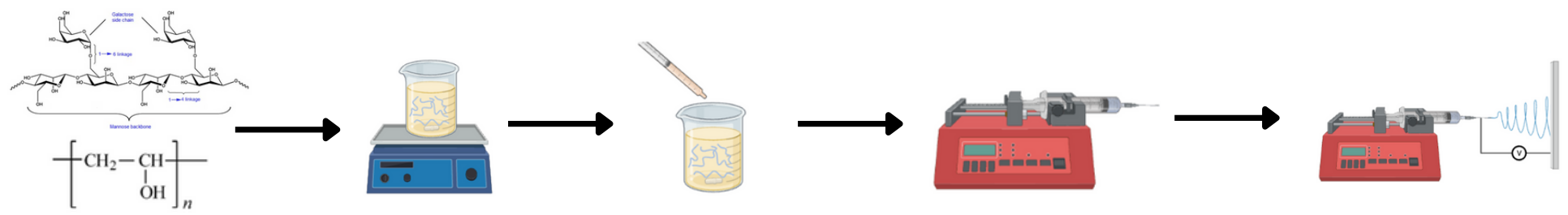
INTRODUCTION

- Chronic wounds and full-thickness skin injuries require advanced materials for effective dermal regeneration.
- Electrospun nanofibers replicate the extracellular matrix, supporting cell growth and healing.
- Fast-dissolving scaffolds enable quick application and integration into moist wound environments [1].
- Polyvinyl alcohol (PVA) offers excellent film-forming and hydrophilic properties.
- Guar gum (GG), a natural polysaccharide, enhances biocompatibility and water retention [4].
- This study investigates PVA/GG nanofiber scaffolds for improved wound healing performance.

OBJECTIVES

- To synthesize PVA/GG nanofibers by varying concentrations of PVA
- To characterize the Physicochemical properties of PVA/GG electrospun nanofibers compared to PVA nanofibers as fast-dissolved scaffolds.

SYNTHESIS OF PVA/GG



METHODOLOGY

- Sample preparation
- Synthesis of PVA NF
- Synthesis of PVA/GG NF
- SEM-EDX
- Dissolution Test
- WVTR

CHARACTERIZATION AND TESTING

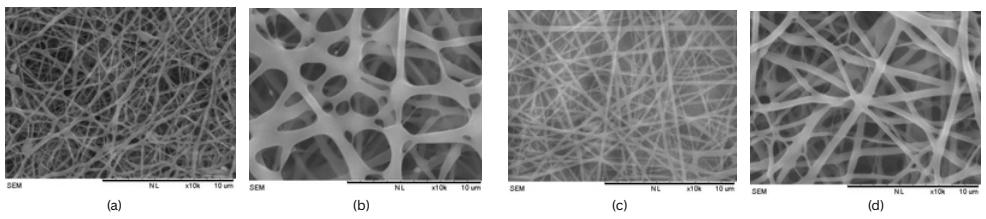


Figure 1. (a) SEM- Pure 10% PVA NF x10k ; (b) SEM- 10% PVA/GG NF x10k ; (c) SEM- Pure 20% PVA NF x10k ; (d) SEM- 20% PVA/GG NF x10k ;

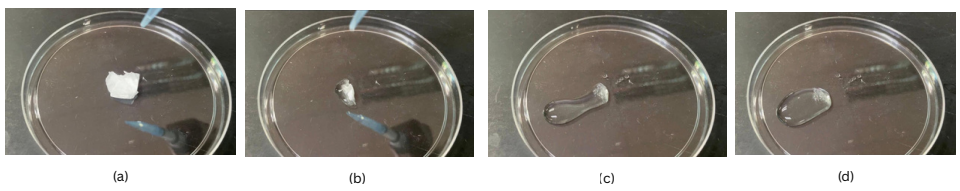


Figure 2. Dissolution Test for 20% PVA/GG (a) 0s ; (b) 5s ; (c) 30s ; (d) 60s ;

Table 1. Elements composition of each membrane

Membrane	Percentage (%)	Element	Weight %	Weight % σ	Atomic %
10% PVA	100	C	66.243	3.601	72.329
		O	33.757	3.601	27.671
20% PVA	100	C	77.987	9.148	82.515
		O	22.013	9.148	17.485
10% PVA/GG	70/30	C	63.061	3.105	69.457
		O	36.939	3.105	30.543
20% PVA/GG	70/30	C	68.08	4.931	73.965
		O	31.92	4.931	26.035

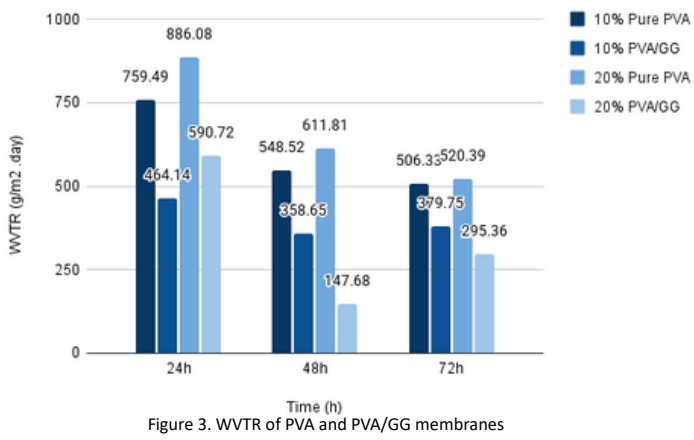


Figure 3. WVTR of PVA and PVA/GG membranes

DISCUSSION

- **Electrospinning Optimization:** Stable, bead-free nanofibers were achieved using 20% PVA or 20% PVA with 1% GG at 0.5 mL/h, 21 kV, and 15 cm distance. Lower concentrations or higher flow rates led to bead formation and irregular morphology.
- **SEM-EDX:** Higher PVA content produced smooth, uniform fibers. Lower concentrations showed bead-on-string morphologies. Fiber diameters ranged from 178–754 nm. EDX confirmed all membranes primarily contained carbon and oxygen, with higher carbon content in 20% PVA (C: 77.99%, O: 22.01%) and increased oxygen in PVA/GG blends (up to 36.94%).
- **Dissolution Test:** 10% PVA/GG dissolved fastest (within 60s), while 20% PVA/GG and PVA alone lasted longer. Faster dissolution was linked to thinner, more porous fibers.
- **Water Vapor Transmission Rate (WVTR):** PVA membranes showed the fastest moisture loss, while PVA/GG blends, especially 10% PVA/GG, maintained more stable WVTR over 72 hours. This indicates that adding GG helps retain moisture, making the blends more suitable for wound environments requiring prolonged hydration.

CONCLUSION

Electrospun PVA and PVA/GG nanofiber membranes were successfully optimized and characterized for wound healing applications. The findings confirm that electrospinning parameters—particularly polymer concentration and flow rate—critically influence fiber morphology, water vapor transmission, and dissolution behavior. Higher PVA content produced uniform, bead-free fibers with better structural integrity, while PVA/GG blends improved moisture retention and controlled dissolution. These tunable properties highlight the potential of electrospun nanofibers to meet specific clinical needs, from rapid drug release to long-term wound protection.

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